GRIDVISION ENERGY KNOWLEDGE HANDBOOK

Commercial & Industrial Energy Efficiency | Knowledge Source: RAG-ABC-001 | Revision 1.0 | 03 Nov 2026

Purpose: A compact, self-contained knowledge source for retrieval-augmented generation (RAG). It provides general principles, definitions, decision guidance and example calculations. It is separate from the factory-specific Operations Register.

1. How industrial electricity costs are formed

An industrial electricity bill can contain energy charges (kWh multiplied by the applicable time-of-use rate), maximum-demand charges (often based on the highest measured demand in a billing period), fixed charges, power-factor-related adjustments, taxes and other utility-specific items. The exact components and calculation rules depend on the local tariff order and the consumer's contract. A GridVision analysis must use the customer's actual tariff schedule and bill rather than assume a universal tariff.

Energy (kWh) measures electricity consumed over time. Demand (kW) measures the rate of electricity use at a point or over a defined demand interval. A facility can reduce kWh while still creating a high demand peak if many large loads operate simultaneously.

2. Load profiles, peak demand and load shifting

A load profile is a time series of electrical demand or energy consumption. Reviewing it by hour, shift, weekday and production state can reveal recurring peaks, baseload consumption, idle loads and unusual changes. Forecasts should be evaluated against a chronological holdout period so that future information does not leak into training.

Load shifting moves a flexible activity from one time window to another without necessarily reducing the total energy required. It can lower cost when the destination window has a lower tariff or when it avoids coincident demand peaks. It is only feasible when production deadlines, machine dependencies, staffing, maintenance and product-quality requirements remain satisfied.

Peak shaving reduces the maximum grid demand during a selected interval. Possible approaches include rescheduling flexible loads, staggering equipment starts, using on-site generation, or discharging a battery. Peak shaving should be assessed against demand-charge rules, battery losses, equipment limits and operational constraints.

3. Practical efficiency opportunities

Compressed-air systems: compressed air is often costly because electricity is converted to compressed air with significant losses. Common actions include repairing leaks, reducing pressure only to the minimum required by end uses, maintaining filters and dryers, and avoiding use of compressed air for tasks that can be performed more efficiently by other means. Pressure changes must be checked against equipment and process requirements.

Motors and pumps: match motor and pump operation to actual process demand. Avoid throttling or bypass practices where a properly engineered variable-speed solution may be more efficient. Maintain alignment, bearings, seals and lubrication. Do not reduce flow below the process minimum.

Cooling and HVAC: maintain heat exchangers and filters, inspect condenser cleanliness, reduce unwanted heat gains, and use approved temperature setpoints. Cooling loads can rise with ambient temperature and production heat. Never compromise temperature, humidity or cooling conditions required for product quality or worker safety.

Furnaces and process heating: reduce avoidable idle time where process rules permit, maintain insulation and seals, review batch scheduling, and recover waste heat only where technically and operationally suitable. Heating recipes, ramp rates, soak times and quality controls must not be altered solely to reduce energy cost.

Lighting: use zone-based controls, occupancy sensors or scheduling where appropriate. Maintain required illumination levels, emergency lighting and safety requirements. Verify that areas are genuinely unoccupied before switching lighting off.

4. Renewable energy and storage

Rooftop solar PV assessment begins with usable roof area, shading, structural suitability, orientation, tilt, local solar resource, grid-interconnection rules and the facility's interval load profile. A preliminary estimate of annual generation is not a substitute for a

site survey or a detailed engineering design.

Self-consumption is the portion of on-site generation used by the facility at the time it is generated. Higher self-consumption can improve the value of solar where export compensation is lower than the retail import rate. Export limits, net-metering or gross-metering rules and utility approvals vary by jurisdiction and must be verified for the site.

Battery energy capacity is measured in kWh; power capability is measured in kW. Battery dispatch decisions should account for usable state-of-charge range, round-trip efficiency, degradation, warranty limits, inverter power, safety systems and the intended use case. A battery may support peak shaving, backup or solar self-consumption, but these uses compete for finite capacity.

5. Simple calculation framework

Metric	Illustrative calculation	Interpretation
Energy charge	Σ (interval kWh × interval tariff)	Use the tariff applicable to each interval
Average demand	Total kWh ÷ hours in period	Average load over the selected period
Load factor	Average demand ÷ peak demand	Higher values generally indicate a flatter load profile
Solar self-consumption ratio	Solar PV energy used on site ÷ total PV generation	Share of generation consumed behind the meter
Simple payback	Initial investment ÷ annual net savings	Screening metric; excludes financing, degradation and replacement
Battery round-trip efficiency	Energy discharged ÷ energy charged	Use measured or manufacturer-supported values

Example (illustrative only): if a process uses 100 kWh during a period priced at ■8/kWh, the energy charge for that consumption is ■800. If the same 100 kWh is shifted to a period priced at ■5/kWh, the theoretical energy-charge reduction is ■300, before considering demand charges, production impacts, losses or other bill components.

6. Optimization principles and constraints

An energy-cost optimizer should minimize the relevant total cost while satisfying production and engineering constraints. Depending on the site, the model may include grid import, time-varying tariffs, demand charges, machine schedules, production targets, renewable generation and battery charge/discharge decisions.

Typical constraints include: machine availability; minimum and maximum operating hours; process sequence and precedence; batch continuity; production deadlines; rated power; minimum utility flow or pressure; maintenance windows; battery state of charge; inverter limits; grid import/export limits; and safety or quality restrictions.

A recommendation is feasible only if all hard constraints are met. If a required constraint or tariff input is missing, the system should state the assumption, request the missing data or abstain from claiming a validated saving. LLM-generated text must not be treated as a substitute for deterministic cost calculations or optimization results.

7. Recommended analysis workflow for GridVision

1) Ingest factory documents and extract machine, process, dependency, constraint and maintenance information with source references.
2) Ingest interval meter data and tariff schedules into structured storage; validate timestamps, units, missing intervals and meter mapping.
3) Build a baseline: energy use, peak demand, load factor, shift-level patterns and current bill estimate.
4) Forecast demand and identify candidate peak windows using chronological evaluation.
5) Retrieve relevant operating constraints and efficiency guidance for the affected assets and processes.
6) Generate candidate actions, then test them with deterministic calculations or a constrained optimizer.
7) Report estimated impact, assumptions, constraints, source references and confidence limitations; require human approval before operational changes.

8. Evidence, provenance and safety

RAG answers should distinguish factory-specific facts from general guidance. Store document title, revision, page, section and ingestion timestamp with each chunk or extracted claim. Prefer current, site-approved documents for site-specific limits. If two documents conflict, surface the conflict and do not silently choose a value. General guidance must never override a factory's approved SOP, applicable law, equipment manufacturer instructions or safety procedure.

The content in this handbook is general educational demonstration material. It is not a tariff interpretation, engineering design, legal opinion, equipment operating procedure or guarantee of savings. Verify current local regulations, utility tariffs, manufacturer documentation and site conditions before acting.

9. Retrieval metadata suggestions

Metadata field	Example value
document_id	RAG-ABC-001
document_type	energy_efficiency_knowledge
topic	compressed_air; load_shifting; solar; battery; optimization
scope	general_industrial_guidance
revision	1.0
effective_date	2026-11-03
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